PA PROJECT

Udemy finance analysis

OCTOBER 11, 2026

Submitted By: Mr. HARSH SINGH
University Roll no.: 1240258188
Submitted To: Mr. DEEPAK SIR

PROJECT: (Predictive analytics)

1. DATA LOADING AND PREPARATION:

Step 1: Data Loading & Cleaning

[1]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns

# Load dataset
df = pd.read_csv("udemy_output_All_Finance__Accounting_p1_p626.csv")

# Handle missing values in price details
df["discount_price__amount"] = df["discount_price__amount"].fillna(
    df["discount_price__amount"].median()
)
df["price_detail__amount"] = df["price_detail__amount"].fillna(
    df["price_detail__amount"].median()
)

# Convert timestamp columns to datetime objects
df["created"] = pd.to_datetime(df["created"])
df["published_time"] = pd.to_datetime(df["published_time"])

print("Cleaned Data Shape:", df.shape ,"\n")

df
```

```
df
```

```
Cleaned Data Shape: (13608, 20)
```

[1]:

	id	title	url	is_paid	num_subscribers	avg_rating	avg_rating_recent	rating	num_reviews	is_wishlisted	nur
0	762616	The Complete SQL Bootcamp 2020: Go from Zero t...	/course/the-complete-sql-bootcamp/	True	295509	4.66019	4.67874	4.67874	78006	False	
1	937678	Tableau 2020 A-Z: Hands-On Tableau Training fo...	/course/tableau10/	True	209070	4.58956	4.60015	4.60015	54581	False	
2	1361790	PMP Exam Prep Seminar - PMBOK Guide 6	/course/pmp-pmbok6-35-pdus/	True	155282	4.59491	4.59326	4.59326	52653	False	
3	648826	The Complete Financial Analyst Course 2020	/course/the-complete-financial-analyst-course/	True	245860	4.54407	4.53772	4.53772	46447	False	
4	637930	An Entire MBA in 1 Course:Award Winning Busine...	/course/an-entire-mba-in-1-courseaward-winning...	True	374836	4.47080	4.47173	4.47173	41630	False	
...	...	...	...	...	...	...	...	...	...	...	...
13603	3005408	מושגים בסיסיים באופציות חלק ב	/course/1-ndgzhu/	False	47	0.00000	0.00000	0.00000	0	False	
13604	3467284	Programa de Integridade	/course/iaudit_programa_de_integridade/	False	19	0.00000	0.00000	0.00000	0	False	
		Goodwill and									

2. FEATURE EXTRACTION:

Step 2: Feature Extraction & Engineering To build robust predictive models, we extract temporal features and engagement ratios from the raw attributes:

[2]:
```python
# Extract temporal features from publication date (with timezone removed)
df["pub_year"] = df["published_time"].dt.year
df["pub_month"] = df["published_time"].dt.month

# Fix: Remove timezone information using .dt.tz_localize(None)
df["course_age_days"] = (
    pd.Timestamp("2026-01-01") - df["published_time"].dt.tz_localize(None)
).dt.days

# Feature Extraction: Review-to-Subscriber Ratio (Engagement Rate)
df["review_sub_ratio"] = df["num_reviews"] / (df["num_subscribers"] + 1)

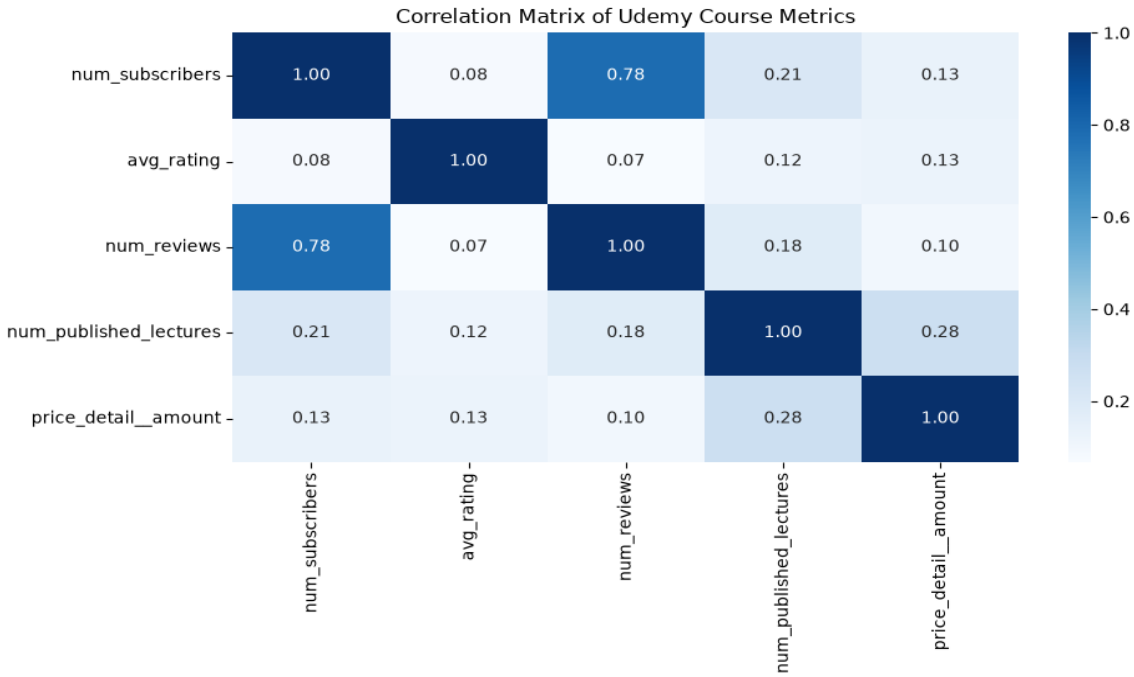
# Feature Extraction: Discount Percentage
df["discount_pct"] = (
    df["price_detail__amount"] - df["discount_price__amount"]
) / df["price_detail__amount"]

# Target Transformation (Log scale for skewed subscriber counts)
df["log_subscribers"] = np.log1p(df["num_subscribers"])
```

3. EDA AND VISUALIZATION:

Step 3: Exploratory Data Analysis & Visualization Python

[3]:
```python
plt.figure(figsize=(10, 6))
sns.heatmap(
    df[
        [
            "num_subscribers",
            "avg_rating",
            "num_reviews",
            "num_published_lectures",
            "price_detail__amount",
        ]
    ].corr(),
    annot=True,
    cmap="Blues",
    fmt=".2f",
)
plt.title("Correlation Matrix of Udemy Course Metrics")
plt.tight_layout()
plt.savefig("correlation_heatmap.png")
plt.show()
```



4. ML MODELING AND EVALUATION:

Machine Learning Model for Future Predictions

[4]:

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

# Define Features and Target
features = [
    "avg_rating",
    "num_published_lectures",
    "num_published_practice_tests",
    "price_detail__amount",
    "discount_pct",
    "course_age_days",
    "is_paid",
]
X = df[features].dropna()
y = df.loc[X.index, "log_subscribers"]

# Train/Test Split
X_train, X_test, y_type_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Initialize and Train Model
model = GradientBoostingRegressor(
    n_estimators=100, learning_rate=0.1, max_depth=5, random_state=42
)
model.fit(X_train, y_type_train)

# Evaluation
y_pred = model.predict(X_test)
print(f"R2 Score: {r2_score(y_test, y_pred):.3f}")
print(f"RMSE: {np.sqrt(mean_squared_error(y_test, y_pred)):.3f}")
```

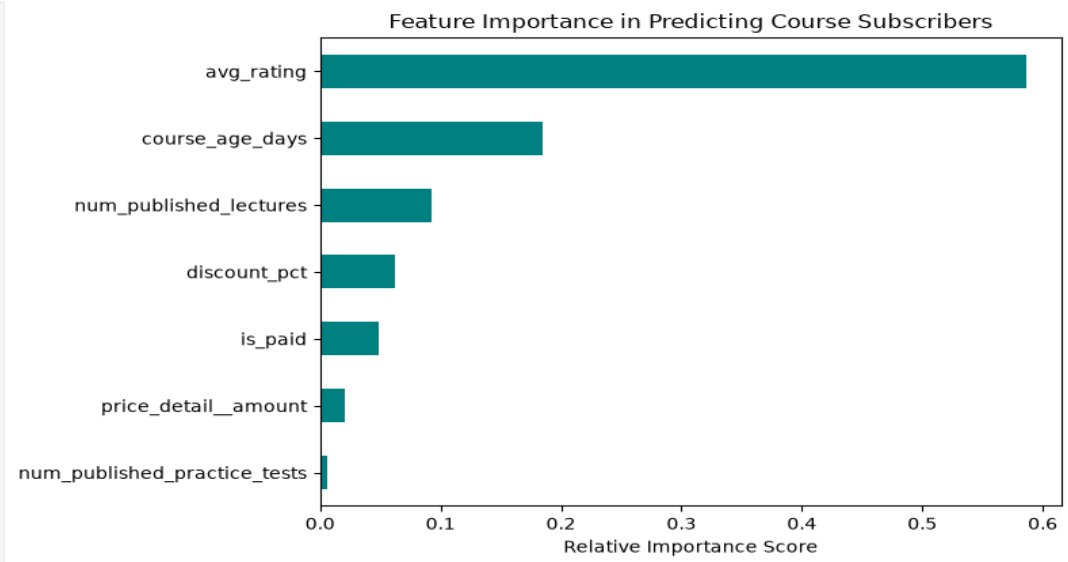
```
R2 Score: 0.471
RMSE: 1.717
```

5. Graphical Presentation & Model Interpretability:

Graphical Presentation & Model Interpretability

[5]:
```
# 1. Feature Importance Plot
plt.figure(figsize=(8, 5))
importances = pd.Series(model.feature_importances_, index=features).sort_values(
    ascending=True
)
importances.plot(kind="barh", color="teal")
plt.title("Feature Importance in Predicting Course Subscribers")
plt.xlabel("Relative Importance Score")
plt.tight_layout()
plt.savefig("feature_importance.png")
plt.show()

# 2. Actual vs Predicted Plot
plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_pred, alpha=0.3, color="purple")
plt.plot(
    [y_test.min(), y_test.max()], [y_test.min(), y_test.max()], "k--", lw=2
)
plt.xlabel("Actual Log Subscribers")
plt.ylabel("Predicted Log Subscribers")
plt.title("Actual vs Predicted Course Success")
plt.tight_layout()
plt.savefig("actual_vs_predicted.png")
plt.show()
```



Collapse Output

